09A — Deep-Dive Feature Audit: Combat Engine & Tactical Simulation (Part 1 of 6)

The Core Damage Math Paradox (`dmg = hpMax(d) * frac`), `ST.explain` vs `ST.get` Split-Brain, Missing Elemental Matchups on Strikes & Unwired Behavioral Mastery Traits

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Mathematically Truthful Xianxia Combat**. Every point of Max HP, Defense, Elemental Affinity, Armor Penetration, and Combat Mastery earned through years of cultivation and crafting must produce authentic, consistent tactical results—both on the character sheet and in live tick-by-tick combat!
- **Master Roadmap for Category 09 (`09A–09F`)**:
  - **`09A` (This Document)**: **Core Damage Formula (`dmg = hpMax(d) * frac`), 12-Stat Composite Engine (`stats.js`), `ST.explain` vs `ST.get` Split-Brain, Elemental Matchup Gaps & Behavioral Mastery Traits (`ACTION_MASTERY_TRAITS`)**
  - `09B`: Spatial Battlefield Geometry, 5-Band Range, `NPC` vs `foe` Movement Asymmetry, Row Bodyblock `24`-Tile Bug, Vanguard Interception & Displacement
  - `09C`: Action Economy, Windup/Cast/Channeling/Poise-Break, Statuses, DoT Self-Harm/Reflect Loops, Acupoints & Boss Anatomy Breaker
  - `09D`: Multi-Entity Units — Summons, Shadow Clones, Constructs, Bonded Pet Assists, Gate Beasts, Weapon Avatars, Ancestral Phantoms & Party Synergies
  - `09E`: Fight Modes, Stakes, Morale/Yield/Retreat, 7-Layer Cheat-Death Stacking, Raid Core-Breach AI, Self-Defense Jail Bug & Heavenly Tribulations
  - `09F`: Grand Finale — Holistic Overview of the Entire Combat Ecosystem (Tick Combat vs Off-Map Abstract `DSH`, 9-Stage Foe Scaling, Mid-Fight Save/Load & Master `P0->P3` Roadmap)

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0. Hard Architectural Invariants

1. **NEVER modify `js/human_core.js`** (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **Preserve `combat_balance.js`, `v240`, `v300` & `v340` Contracts**: Equal-stage baseline TTK (`~8–12` exchanges) and cross-stage dominance (`+1` stage $\ge 80%$ win rate) must remain crisp while eliminating perverse stat inversions!

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1. [CMB-MATH-01] Critical Core-Formula Paradox (js/combat.js:533): Multiplying Incoming Damage by Defender's Own C.hpMax(d) Cancels 100% of +HP Survivability Against Strikes & Cripples Flat Healing!

1.1 Exact Source Proof (js/combat.js:445–533)

Inspect the core strike damage equation in COMBAT.strike:

```javascript
let frac = 0.2 * (A0 / (A0 + 0.8 * Math.max(0, D))) * (o.M || 1) * (o.synDmgMult
|| 1) * ...;
...
const maxSingleHitFrac = (dEqStage >= 5 && !o.forced) ? 0.48 : 1.85;
frac = clamp(frac, o.forced ? 0.0001 : 0.005, maxSingleHitFrac);
if (nonlethal) frac *= 0.55;
const hm = C.hpMax(d), ..., dmg = hm * frac * beastGuard;
```

1.2 Four Severe Multi-Order Chain-Reaction Distortions:

1. **Equipping +HP Gear or Mantras Linearly Increases Incoming Damage Taken (Zero Extra Hits Survived!)**:
   - Notice frac depends **only** on A0 and D—it does **not** depend on C.hpMax(d). Then dmg = C.hpMax(d) * frac multiplies frac by the **defender's own Max HP**!
   - **Concrete Proof**:
     - Disciple with hpMax = 1,000, def = 200 hit by Bandit (A0 = 200): frac = 0.1111 $\to$ takes **111 damage** (dies in 9.0 hits).
     - Same Disciple equips Legendary Heavy Armor + Longevity Mantra raising hpMax to **3,840 HP** (+284% HP!): frac is still 0.1111 $\to$ the exact same Bandit now hits for **427 damage per swing** (3,840 / 427 = STILL dies in 9.0 hits!!)!
     - **Every +HP bonus in the game (BODY path weight, HEAVY armor, NECKLACE, EVERGREEN mantra, PEAK HP bonuses, Constitution traits) adds 0.00 extra hits of survival against normal & skill strikes!**
2. **High Max HP Actively Punishes Flat Potions/Heals & Feeds Enemy Lifesteal/Reflect**:
   - Because the 3,840-HP tank takes 3.84× larger raw dmg numbers (427 instead of 111), a flat P:BASIC_HP (+42 HP) or EFX_MASS_VITAL_HEAL (+48 HP)

restores `3.84×` fewer hits of health, **while a `DEMON` enemy with `15%` lifesteal heals `+64` HP per hit off the tank instead of `+16` HP off a squishy mage**!

3. **`0.5% MaxHP` Minimum Damage Floor (`frac = clamp(frac, 0.005, ...)`) Lets Stage-1 Mortals Chip Down Stage-9 Gods**:
   - Even when `A0 = 12` vs `D = 5,000` (`raw ratio = 0.0006`), `clamp(frac, 0.005, ...)` forces every grazing hit to deal at least `0.5%` of the Stage-9 Patriarch's `80,000` Max HP (`400` damage per pebble)!
4. **Split Reality Between `STATS.rating` (`Off-Map Combat`) and `COMBAT.strike` (`On-Map Combat`)**:
   - `STATS.rating` (`stats.js`, used by `CULT.power` in Sieges, Sanctums, Pagoda, and `DSH`) computes `hpE = v.hp * ...` assuming `v.hp` scales EHP linearly—causing high-HP builds to overperform in abstract off-map bouts while collapsing in live tick-by-tick fights!

1.3 Clean Calibration Fix (`Reference-HP Anchor Preserving Exact Baseline TTK!`)

Instead of multiplying `frac` by `C.hpMax(d)`, multiply `frac` by the **defender's unbuffed stage-baseline reference HP (`CULT.baseStageHp(d)` — or `Math.sqrt(C.hpMax(d) * refStageHp) / A0-anchored` reference `refHp = CULT.stageRefHp(dEqStage)`)**:

- When a character has standard baseline HP (`hpMax === refHp`), `dmg = refHp * frac` produces **100% identical damage numbers to current `combat_balance.js` (`preserving ~9-hit baseline TTK`)**!
- When a tank invests in `+100%` Max HP (`hpMax = 2.0 * refHp`), incoming Bandit damage stays `refHp * frac` (`111` dmg instead of `222`!), so **`+100%` Max HP genuinely doubles hits-to-die (`9 -> 18` hits!)**, makes flat heals meaningful, stops feeding enemy lifesteal, and aligns live combat with `STATS.rating`!
- Also lower the minimum `frac` floor when `aEqStage <= dEqStage - 3` (`realm suppression floor: 0.0008` instead of `0.005`).

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2. `[CMB-STAT-02]` Five Divergences & Unwired Modifiers in the 12-Stat Composite Engine (`js/stats.js` & `js/combat.js`)!

1. **`ST.compute` (`UI Character Sheet`) vs `ST.getBaseFast` (`Actual Combat`) Split-Brain (`js/stats.js`)**:
   - **`MECH_PUPPET` (`stats.js:314` vs `534`)**: `ms.integrity || 100` treats `0` broken integrity as `100%` full integrity, and `mode === 'OVERDRIVE'` displays `+8% ATK/MATK, +8 PEN` in `explain` while falling into `else` (`+18% DEF, +0% ATK`) in `getBaseFast` (`01D`)!

- **KARMA_SAMSARA (stats.js:207, 320 vs 545)**: passiveCacheKey hashes only stance.charCodeAt(0) ('ASURA' === 'ANIMAL' === 65), and all 6 realm stances disagree between compute and getBaseFast!
- **31 Grand Dao subpathTraits (stats.js:503)**: c.track starting at ordinal = 1 shadows c.stage = 8 in getBaseFast, locking high-stage elders out of their subpath traits!
- **Fix**: Extract a single shared pure helper ST.applyPathAndSubpathModifiers(n, p, f, parts) called by **both** compute and getBaseFast!

2. **Broken Low-HP Conditional Passive Denominator (n._hpf in js/stats.js)**:
   - n._hpf divides n.h.state.hp by raw hpMaxBase (*before* armor/mantra/passive +HP%) instead of CULT.hpMax(n), causing heavily geared tanks at 45% actual HP to read _hpf = 1.0 (never triggering low-HP passives!), while -hp% blood-cultivators at 100% full HP read _hpf = 0.80 (permanently injured)!
3. **Ghost it.combatMult & it.defMult Equipment Fields (05E)**:
   - Stored and upgraded across economy.js and scored in M.gearSuitability, yet **never applied in stats.js (0% effect)**! Wire p.atk += it.combatMult * df; p.matk += it.combatMult * df; p.def += it.defMult * df; p.mdef += it.defMult * df; cleanly!
4. **Two Half-Wired ACTION_MASTERY_TRAITS (js/combat.js:1218–1233 vs 662–667)!**:
   - **TRAIT_VANGUARD_GUARD** (combat.js:1221) defines allyDefAuraPct: 0.12 (*"ป้องกันตนเองและเพื่อนข้างเคียง +12%"*), yet **only** self defBonusPct is applied—nearby allies receive **0%**! Apply 0.88× incoming damage (or +12% def) to allies within 2.8 tiles of a living TRAIT_VANGUARD_GUARD vanguard!
   - **TRAIT_FORMATION_STRATEGIST** (combat.js:1228) defines statKey: 'formationTicks', threshold: 30, qiRegenBonusPct: 0.15—**yet recordAction(W, u, 'formationTicks', 1) is NEVER called anywhere in combat.js during live combat (0 organic unlocks!), AND C.fightStep (lines 662–667) NEVER multiplies qRegen by 1 + mods.qiRegenBonusPct (0% Qi regen effect even when unlocked!)**!
     - **Fix**: Increment formationTicks in C.fightStep / tickDomain when an ARRAY_TALISMAN / STRATEGY_GOVERNANCE / domain-deploying cultivator maintains an active formation/domain in serious combat, and multiply qRegen in C.fightStep (line 666) by (1 + (C.ACTION_MASTERY_TRAITS.hasTrait(n, 'TRAIT_FORMATION_STRATEGIST') ? 0.15 : 0))!

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3. [CMB-ELEM-03] Basic Attacks, Counters, Pet Assists & Relic Strikes Bypass the Entire Elemental Counter Matrix (js/combat.js:480 vs 728)!

- **Exact Code Proof**:
  - `ELEM.resolveCombatInteraction(a, d, p)` is called **only inside `useSkill` (`line 728`)**, passing `synDmgMult` into `C.strike`!
  - Whenever `C.strike` is called directly—**basic weapon attacks (`line 622`), `counter` / `astralMirage` ripostes (`lines 461, 470`), ZoC opportunity strikes (`line 218`), `triggerPathGaugeFinisher` (`v340:59`), and elemental beasts/weapons**—`o.synDmgMult` is `undefined` (`1.0`), so a **Fire Spirit Beast or Fire-Runed Blade hitting a Water/Wood target gets `0%` elemental matchup modifier**!
- **Exact Fix**: Inside `C.strike`, when `o.synDmgMult == null` and `g.ELEM && g.ELEM.combatElementalMatchup`, compute lightweight weapon/innate elemental matchup (`atkElem` from `o.skill?.element || a.equip?.WEAPON?.element || a.element` vs defender `d`) so elemental weapons, runes (`E05`), and elemental beasts always honor the 10-Element counter wheel!

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4. Mandatory Verification Suite Additions (09A)

1. **`+HP` True Survivability Invariant Test**: Measure hits-to-die for an NPC at `1,000` Max HP vs the same NPC with `+100%` Max HP (`2,000` HP, identical `def`) against the same attacker; assert the `2,000`-HP build survives `~2.0×` as many hits!
2. **`ST.explain` === `ST.get` Parity Sweep Across All 43 Paths & Modes**: Iterate all 43 paths (including `MECH_PUPPET 'OVERDRIVE'` / `integrity = 0` and all 6 `KARMA_SAMSARA` stances) and assert `JSON.stringify(STATS.explain(n).v) === JSON.stringify(STATS.get(n))`!
3. **Live `TRAIT_FORMATION_STRATEGIST` Unlock & Qi Regen Test**: Deploy a battlefield domain for 30 ticks in live `COMBAT.tick`, assert `TRAIT_FORMATION_STRATEGIST` unlocks organically, and assert `C.fightStep` Qi regen increases by `+15%`.